Abhishek Ningappa Walikar

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Profile

M.Sc. Computer Science student at Universität Paderborn with professional experience in Python, machine learning, data workflows, and software development. Hands-on work spans Transformer fine-tuning, deep-learning evaluation, explainable graph ML, and an end-to-end LLM application with structured outputs, retrieval-augmented generation, agentic validation, automated testing, and deployment. Experienced with PyTorch, TensorFlow, scikit-learn, Git/Linux, CI/CD, APIs, and reproducible experimentation. Seeking Working Student or internship opportunities in Applied AI, Machine Learning, AI Engineering, NLP/LLMs, Computer Vision, Graph ML, and Data Science.

Education

04/2025 – Present **M.Sc. in Computer Science, Universität Paderborn**
Paderborn, Germany | Expected 03/2028 | Current Grade: 2.6
- Relevant coursework: Explainable Artificial Intelligence, Machine Learning for Face Biometrics, Data-Driven Engineering (1.3), Data-Driven Innovation (2.0), Foundations of Knowledge Graphs (current).

01/2021 – 08/2022 **Master of Computer Applications, Karnatak University**
India

08/2017 – 10/2020 **B.Sc. Computer Science, JSS SMI UG & PG Studies**
India

Professional Experience

04/2023 – 12/2024 **Software Engineer – Python, Data & ML**
CABCS Group India Technologies Pvt. Ltd., Bangalore, India
- Developed and maintained Python-based software and data-processing workflows for extraction, transformation, cleansing, validation, and preparation across multiple datasets, supporting repeatable analytics and ML workloads.
- Fine-tuned and evaluated a Hugging Face Transformer with PyTorch for automated text classification, achieving **96%+ classification accuracy** on a self-collected and curated text dataset.
- Built reusable ML experimentation pipelines with scikit-learn and TensorFlow across multiple datasets and model configurations, supporting structured validation, benchmarking, and reproducible performance analysis.

08/2022 – 02/2023 **Full-Stack Developer Intern**
KodNest India Pvt. Ltd., Bangalore, India
- Developed 10+ Python application features and integrated and tested 6+ REST APIs, gaining practical experience with backend logic, databases, application testing, and reliable data flow.

Key Projects

04/2026 – 10/2026 **Explainable Knowledge-Graph Classification**
Machine Learning, Explainability & Graph ML Project
GitHub Repository
- Developed a leakage-safe explainable classification pipeline on a knowledge graph, removing 79 direct label-leakage edges and replacing identity-heavy one-hot inputs with **662 readable graph features** evaluated with nested repeated cross-validation.
- Improved an interpretable decision tree from **63.4% to 86.2%** nested-CV accuracy, while LightGBM reached **89.8% nested CV and 91.7% test accuracy**. Cross-checked explanations with multiple complementary methods and an 83.3% fidelity surrogate.
- Benchmarked interpretable models against a graph neural network and a second dataset, then packaged the workflow with **45 unit tests plus a 5-test model-quality gate**, a one-command pipeline, and automated CI.

10/2025 – 08/2026 **Machine Unlearning for Face Recognition – Model Evaluation & Reliability**
Research Project | Deep Learning & Computer Vision | Universität Paderborn
- Co-developed an adversarial-probing evaluation method for machine unlearning in face-recognition models, revealing representation changes that standard clean-input evaluation may miss.
- Evaluated the method on a **5.82M-image dataset** across 3 machine-unlearning approaches and 7 model configurations, producing stronger separation in **40 of 41** completed single-identity configurations.
- Benchmarked multiple machine-unlearning approaches across different face-recognition models and forgetting scales from 1 to 1,000, comparing evaluation behavior across varied unlearning settings.
- Optimized adversarial-search settings through systematic experiments, reducing full-run runtime from 1,215 s to 690 s, approximately **43%**, while changing the evaluation score by less than 0.001.

01/2025 – 04/2026 **LLM-Based System Context Analysis Assistant**
Applied LLM & Software Engineering Project | Universität Paderborn
GitHub Repository | Live Demo
- Developed a Python LLM application with structured outputs, schema and referential-integrity validation, custom embedding-based retrieval, agentic self-validation, and 3 interactive graph views.
- Built a 6-case hand-labeled evaluation harness achieving **0.88 Entity F1**, **0.82 Relationship F1**, 94% risk coverage, and 100% evidence grounding. The optional agentic loop improved Entity F1 from 0.879 to 0.924, with the measured latency and cost trade-off documented.
- Hardened and deployed the application with **69 pytest unit tests**, zero network calls in the test suite, Docker, a 2-job GitHub Actions CI/CD pipeline, Infrastructure-as-Code configuration, and a live Render deployment.

Technical Skills

Programming / Data	Python, SQL, pandas, NumPy, SciPy
ML / Deep Learning	PyTorch, TensorFlow, scikit-learn, LightGBM, Hugging Face Transformers
LLM / AI Systems	LLM API integration, RAG, LangGraph, structured outputs, Pydantic
Graph ML / XAI	rdflib, PyTorch Geometric, R-GCN, SHAP, nested cross-validation
Software Engineering	pytest, REST APIs, Docker, GitHub Actions CI/CD, Git/GitHub, Linux

Languages

English	Fluent, C1
German	A2 completed, learning B1